

SKILL 4

To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

```
#include <stdio.h>
#include <string.h>

int main() {
    char history [5][100];
    char command [100];
    int count = 0;

    while (count < 5) {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);
        command[strcspn(command, "\n")] = '\0';
        if (strcmp(command, "exit") == 0)
            break;

        strcpy(history[count], command);
        count++;
    }

    printf("\nCommand History:\n");

    for (int i = 0; i < count; i++) {
        printf("%d: %s\n", i + 1, history[i]);
    }
    return 0;
}
```

Output:

```
shashank@LAPTOP-24KPR0R8:~$ nano Cmdhistory.c
shashank@LAPTOP-24KPR0R8:~$ gcc Cmdhistory.c -o Cmdhistory
shashank@LAPTOP-24KPR0R8:~$ ./Cmdhistory
myshell> ls
myshell> pwd
myshell> exit

Command History:
1: ls
2: pwd
```

b)

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Node {
char command [100];
struct Node *next;
};

int main() {
struct Node *head = NULL;
struct Node *newNode;

newNode = malloc(sizeof(struct Node));

if (newNode == NULL) {
printf("Memory allocation failed\n");
return 1;
}
strcpy(newNode->command, "ls");
newNode->next = head;
head = newNode;

printf("Command: %s\n", head->command);

free(head);

return 0;
}
```

Output:

```
shashank@LAPTOP-24KPROR8:~$ gcc dynamic.c -o dynamic
shashank@LAPTOP-24KPROR8:~$ ./dynamic
Command: ls
```